Aditya Peer

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Research Interest

Interested in systems and networking research, with a focus on hardware–software co-design frameworks for AI and data center infrastructure. Broad research interests include programmable data planes and scalable distributed systems.

Education

- **Indraprastha Institute of Information Technology Delhi** — New Delhi, India
  Ph.D (Computer Science & Engg.) — *July 2025 - Present*
  - Advisor: Dr. Rinku Shah
  - Research Topic: Re-designing Distributed Inference Networking for Next-Generation Scale-Up AI Infrastructure
  - Current CGPA: 9.22
- **Indraprastha Institute of Information Technology Delhi** — New Delhi, India
  Master of Technology - Dual Degree (Computer Science) — *2020 - 2025*
  - CGPA: 8.14
- **Amity International School** — Uttar Pradesh, India
  CBSE, Standard 12, PCM — *2018 - 2020*
  - Percentage: 91.2

Publications & Patents

1. **Aditya Peer**, Rohan Sudhir Basugade, Siddhant, Neeraj Kumar Yadav, Agamdeep Singh, Praveen Tamanna, Sumit Darak, Rinku Shah. "FlexMesh: Designing Size-aware Cryptographic Primitives for FPGA-based Accelerators" , In Proceedings of the 22nd ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT'26), Utrecht, The Netherlands.
   (CORE-A, among Top-3 Networking Conferences)
2. **Vedant Bothra, Aditya Peer**, Vijay Singh, Mukulika Maity and Rinku Shah. "CoDel-ACT: Realizing CoDel AQM for Programmable Switch ASIC" in the International Federation for Information Processing (IFIP) Networking 2024 held in Thessaloniki, Greece, 2024.
   (CORE-B ranked conference)

Skills

- **Programming Languages:** Python, C/C++, HLS, P4, MATLAB
- **Libraries/Software Packages**: numpy, pandas, pytorch
- **Other Softwares/Tools**: Linux, Git, ns3, Vitis-2022.1
- **Programmable Devices**: Tofino switch, FPGAs- Zedboard, Zcu706.

Awards & Medals

- Joined the NSDI'26 Artifact Evaluation Committee
- Received the Best Poster Award at IndoSys'26 for the paper "Designing Size-Aware Cryptographic Primitives for FPGA-Based Accelerators"
- Presented in top-5 talks RIISE 25, IIIT Delhi, for the work on building size-aware Cryptographic Primitives for FPGA-based Accelerators.
- CoNEXT student travel grant to attend CoNEXT 2025, physically held in HKUST, Hong Kong.
- M.Tech thesis award on the Project: Designing Size-aware Cryptographic Primitives for FPGA-based Accelerators.
- SURF Excellence Award on the Project: Realizing CoDel AQM for Programmable Switch ASIC, Summer [2023]
- Completion of Data Structure and Algorithms course from Coding Ninjas

Projects

- **Designing Size-aware Cryptographic Primitives for FPGA-based Accelerators**
  Research Project
  Telecom operators and military/defense applications (e.g., UAVs) prefer to offload compute-intensive cryptographic processing to FPGA-based cryptographic accelerators over fixed-function ASICs for reconfigurability and short deployment cycles. The state-of-the-art FPGA-based cryptographic accelerators are designed either for high throughput or power efficiency. We propose FlexMesh, a programmable, dynamically reconfigurable, and size-aware framework for 5G/6G cryptographic primitives using FPGAs. By leveraging FPGA adaptability, our approach supports emerging algorithms and optimizes performance across dynamic workloads, helping future-proof cryptographic acceleration for next-generation networks. [Paper] [Project Repo]
- **Real-time analysis of packet queue delays in PDPs (C++)**
  Research Project
  The objective of this project is to implement AQM algorithms (e.g Codel) over Intel Tofino switch. Fine-grained queue measurements at the switches have a wide range of use-cases such as stopping congestion-related attacks, avoiding conflicting workloads, and deploying AQM (Active Queue Management) schemes. The goal of an AQM technique is to manage switch/router buffers in order to reduce network congestion and/or improve end-to-end latency. [Paper]
- **Distributed Systems Concept and Design**
  Undergraduate Project
  In this course, we learned about different RPC frameworks such as gRPC, RabbitMQ, and ZeroMQ. We implemented K-Means using MapReduce from scratch using gRPC. We also implemented Raft Consensus Modified Algorithm System by introducing lease concept to make a distributed system ensuring fault tolerance and consistency. [Project Repo]

Professional Experience

- **Department of Computer Science & Engg.** New Delhi, India
  Indraprastha Institute of Information Technology Delhi *July 2025 to Dec 2025*
  - *Teaching Assistant (Aug 2025 to Dec 2025)*
    Worked as Teaching Assistant for the course Operating Systems. The work involves grading answer sheets, providing tutorials, and remedial lessons to students.

Interests and Hobbies

- I have an interest in Coding to develop my logic skills stronger and love to play chess.
- Playing Cricket, cycling, Kabaddi, football, basketball, and listening to music.

References

- Dr. Rinku Shah
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